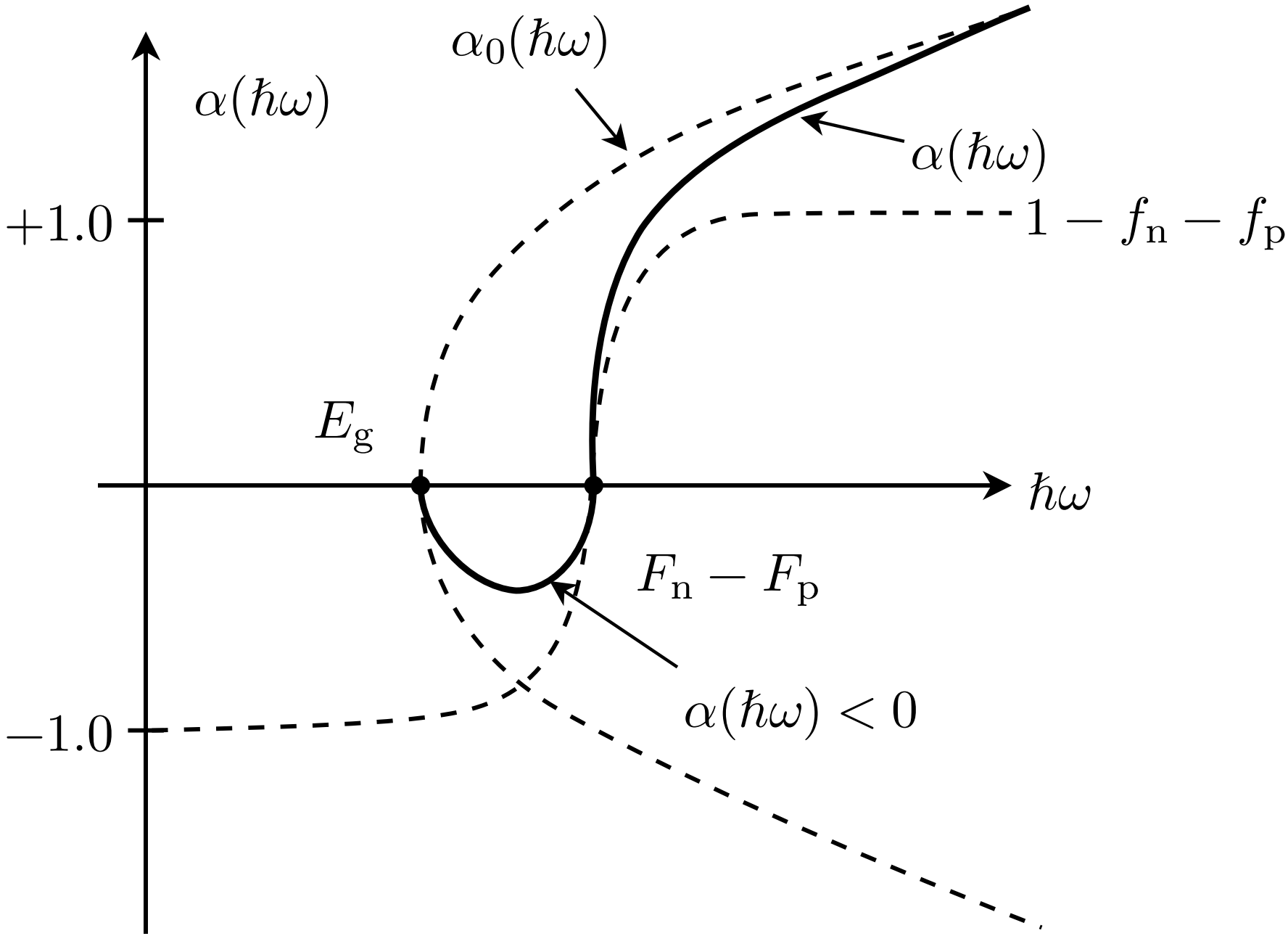
$\alpha_0(\hbar\omega)$
$\alpha(\hbar\omega)$
$\alpha(\hbar\omega)$
$+1.0$
$1 - f_{\mathrm{n}} - f_{\mathrm{p}}$
E_{g}
$\hbar\omega$
$F_{\mathrm{n}} - F_{\mathrm{p}}$
$\alpha(\hbar\omega) < 0$
-1.0